Multilevel machine learning

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1 Introduction

Multilevel (hierarchical) data, in which lower-level units are nested within higher-level units, are ubiquitous—for example, children in schools or patients in hospitals. Units belonging to the same group typically share contextual conditions and latent characteristics, which makes them more similar to each other than to units belonging to different groups. This poses challenges, since many commonly used statistical and machine learning methods assume that observations are independent and identically distributed, an assumption that does not hold in the presence of clustering (Snijders et al., 2012). Ignoring this structure in statistical analysis violates independence assumptions and can lead to misleading results (Wu, 2006; Liang et al., 1986). When within-group dependence is overlooked, models tend to underestimate uncertainty, overstate the amount of information in the data, and learn patterns that reflect group-specific systematic differences rather than generalise relationships (Snijders et al., 2012; Shahbazi et al., 2025). These issues can be amplified in high-dimensional settings, where flexible machine learning methods such as decision trees or random forests are often preferred but still assume independent observations (Hastie et al., 2009; Hajjem et al., 2011).

Traditional approaches to hierarchical data rely on linear and generalised linear mixed-effects models, which account for clustering through random effects and provide valid inference under appropriate assumptions (Gelman et al., 2006; Snijders et al., 2012). Extensions incorporating regularisation, such as mixed-model LASSO variants, have been proposed to address high-dimensional settings (Schelldorfer et al., 2014; Yang et al., 2023). By contrast, many machine learning methods—including random forests and neural networks—typically assume i.i.d. observations (James et al., 2021; Hastie et al., 2009). To bridge these frameworks, Hajjem et al. (2011) introduced an EM-based algorithm to estimate random effects conventionally while using regression trees or random forests for the fixed part for linear regressions. This was later extended to generalised linear regressions (Hajjem et al., 2017). Related EM-derived tree methods include RE-EM trees (Sela et al., 2012) and the M-CART (Lin et al., 2019). Other multilevel tree methods which do not rely on EM have been developed, such as GLMM trees (Fokkema et al., 2018), GMET trees (Fontana et al., 2021) and approaches based on Bayesian or probabilistic trees (Speiser et al., 2020; Wundervald et al., 2023). These ideas generalise to forests through MERF (Mixed effect random forest) and its successors (Hajjem et al., 2014; Ngufor et al., 2019; Pellagatti et al., 2021; Bergonzoli et al., 2024), as well as Bayesian forest extensions (Speiser et al., 2019). Boosting-based mixed-effects models have also been proposed (Sigrist, 2022, 2024). Beyond tree-based algorithms, few multilevel ML methods exist, with neural-network adaptations representing a rare exception (Crane-Droesch, 2017; Zhang et al., 2021).

Despite recent multilevel extensions of machine learning, not much has been done to systematically compare these models. The consequences of applying standard machine learning algorithms to clustered data remain only partly understood, especially for predictive accuracy and variable-importance reliability (Hastie et al., 2009; Register, 2025). Comparative studies between classical mixed-effects models and machine learning methods that do or do not account for clustering are limited and often restricted to narrow applications (Shahbazi et al., 2025; Register, 2025; D'Ottaviano et al., 2022; Mangino et al., 2021). Moreover, the robustness of existing approaches to violations of assumptions about fixed effects, random effects, or the data-generating process is not well established. In high-dimensional settings is also poorly understood, particularly when clustering interacts with sparsity, multicollinearity or complex nonlinear relationships. Consequently, we still lack a clear picture of how these methods respond to hierarchical structure and what types of inferential or modelling errors arise when it is ignored.

In light of these limitations, the study investigates *under which condition a multilevel machine learning model outperform a multilevel regression model, and also what is the effect of clustered data on the performances of machine learning models.* As a reference I also include standard regression models and their multilevel expantion. The study focus on binary classification The rest of report is organised in two main sections. The methodology section describes the modelling framework: logistic regression, multilevel logistic regression, tree-based models, GMERT/GMERF, the simulation design, and evaluation metrics. The simulation section presents results across linear, nonlinear, interaction, multicollinearity, and high-dimensional scenarios, followed by potential extensions.

2 Methodology

To address the research question, I conduct a simulation study comparing single-level and multilevel versions of different modelling approaches. The analysis focuses on binary outcomes and considers four main families of methods: generalized linear models (logistic regression), generalized linear multilevel models (logistic mixed-effects regression), tree-based algorithms (classification trees and random forests), and their multilevel extensions (GMERT and GMERF). Each method is evaluated across a set of controlled data-generating mechanisms that vary the strength and form of the hierarchical structure as well as the dimensionality of the predictor space. This design makes it possible to compare models under comparable conditions and to examine how explicitly modelling the multilevel structure affects their behaviour across different scenarios.

2.1 Logistic Regression

The first technique considered is logistic regression, which provides the baseline single-level model for binary outcomes $y_i \in \{0, 1\}$. It assumes that all observations are independent and relates the probability of success to a linear predictor

$$\eta_i = x_i^\top \beta,$$

where x_i is a for each element i a column vector of p features, and β the p regression coefficients. through the logit link

$$Pr(y_i = 1 \mid x_i) = p_i = \text{logit}^{-1}(\eta_i)$$

where $\text{logit}(p) = \log(\frac{p}{1-p})$ and $\text{logit}^{-1}(\eta_i) = \frac{1}{1+\exp(\eta_i)}$ Under this formulation,

$$y_i \sim \text{Bernoulli}(p_i)$$

and parameter estimation is carried out by maximizing the binomial log-likelihood with respect to β (Gelman et al., 2006). Logistic regression serves as a natural benchmark: it ignores clustering entirely, allowing comparison with methods that explicitly incorporate the hierarchical structure.

2.2 Generalised Linear Multilevel Model

To account for clustering, logistic regression can be extended by introducing group-specific random effects. For observation i in cluster j, the linear predictor becomes

$$\eta_{ij} = x_{ij}^\top \beta + z_{ij}^\top b_j,$$

where $b_j \sim \mathcal{N}(0_p, \mathbf{D})$ is a vector of random effects for cluster j, such as a random intercept or random slopes, capturing how that cluster deviates from the population-level coefficients. The vector z_{ij} is the corresponding design vector indicating which random effects apply to observation i; for instance, $z_{ij} = 1$ in a random-intercept model, or $z_{ij} = (1, x_{ij}))$ when both a random intercept and slope are included.

The outcome is modelled as

$$y_{ij} \sim \text{Bernoulli}\big(\text{logit}^{-1}(\eta_{ij})\big)\,,$$

and inference targets both the fixed effects $\boldsymbol{\beta}$ and the covariance matrix $\mathbf{D}$ governing the distribution of the random effects. Because the likelihood involves integrating over b_j, it lacks a closed-form expression; estimation therefore relies on approximations such as Laplace or adaptive Gaussian quadrature, which provide maximum-likelihood (or restricted maximum-likelihood) estimates under the assumed hierarchical structure (Gelman et al., 2006)

Logistic mixed-effects models offer a principled way to model within-cluster dependence and form the classical benchmark for hierarchical binary data.

2.3 Tree-based techniques

Classification trees partition the predictor space into terminal regions $R_1, \ldots, R_T$ by recursively splitting the data. At each step, the algorithm selects the predictor and cutpoint that best improve node purity, typically measured by the Gini index

$$G(R) = \hat{p}_R(1 - \hat{p}_R),$$

where $\hat{p}_R$ is the proportion of observations with $y = 1$ in region R. The predicted probability for any x in a terminal node is simply $\hat{p}_R$ (Hastie et al., 2009).

Random forests extend trees by averaging predictions across many bootstrapped trees, each grown using random subsets of predictors and a random sample of the with replacement. For a new observation x, the forest estimate is

$$\hat{p}(x) = \frac{1}{B} \sum_{b=1}^{B} \hat{p}_{T_b}(x),$$

which reduces variance and yields more stable predictions.

Both trees and forests model nonlinearities and interactions without distributional assumptions but treat observations as independent (Hastie et al., 2009). They therefore form the single-level machine-learning baselines for comparison with multilevel extensions such as GMERT and GMERF.

2.4 GMERT and GMERF

Multilevel extensions of tree-based methods aim to embed the hierarchical structure of generalized linear mixed-effects models within a non-parametric framework. In a GLMM for binary outcomes, the linear predictor can be written as

$$\eta_{ij} = \underbrace{f(x_{ij})}_{\textit{fixed part}} + \underbrace{z_{ij}^\top b_j}_{\textit{random part}},$$

where $f(x_{ij})$ represents the fixed part, describing the population-level relationship between predictors and the response, and $b_j \sim N(0, D)$ represents the random part, modelling cluster-specific deviations. Standard GLMMs restrict the fixed part to the parametric form $f(x_{ij}) = x_{ij}^\top \beta$. GMERT (Generalised Mixed Effect Regression Tree) replaces this linear component with a regression tree, allowing the fixed part to take an arbitrary piecewise-constant structure while preserving the usual random-effects representation.

To estimate the model, Hajjem et all. (2017) adapt the penalized quasi-likelihood (PQL) algorithm used in GLMMs. At each iteration, the model is linearized around the current fitted values, yielding a pseudo-response y_{ij}^* and working weights. Conditioned on the current estimate of the tree, the random effects are updated through the mixed-model equations that arise from the approximate linear model

$$y_{ij}^* = f^{(t)}(x_{ij}) + z_{ij}^\top b_j + \varepsilon_{ij},$$

treating the fixed part as known. Once the random effects are updated, the fixed part is re-estimated by fitting a weighted regression tree to the adjusted data $y_{ij}^* - z_{ij}^\top b_j^{(t+1)}$, in the same spirit as the MERT algorithm for continuous outcomes. Variance components such as D are then updated using the closed-form expressions derived from the mixed-effects approximation. These steps—PQL linearization, random-effects update, tree refitting, variance-component update—are iterated until the linear predictor converges.

This procedure contrasts with standard GLMM inference, where the fixed effects are obtained through parametric likelihood optimization (e.g., Laplace approximation or adaptive Gaussian quadrature) (Breslow

et al., 1993). In GMERT, the optimization step for the fixed part is replaced entirely by recursive partitioning applied to the PQL-adjusted data, meaning that the structure of $f(\cdot)$ is learned non-parametrically at every iteration rather than being determined by a pre-specified linear predictor.

By embedding a regression tree inside the mixed-effects estimation cycle, GMERT produces a model that retains the hierarchical interpretation of a GLMM while allowing the fixed part to adapt flexibly to nonlinearities and interactions. GMERF (Generalised Mixed Effect Random Forest) follows the same logic, replacing the tree step by a random forest to further stabilize and regularize the non-parametric fixed-effects estimate (Hajjem et al., 2017; Pellagatti et al., 2021).

2.5 Simulating data

To evaluate the four modelling approaches under controlled conditions, I generate synthetic clustered datasets based on a logistic mixed-effects structure. For observation i in cluster j, outcomes follow

$$y_{ij} \sim \text{Bernoulli}\big(\text{logit}^{-1}(f(x_{ij}) + b_{0j} + b_{1j}x_{ij,1})\big)\,,$$

with $(b_{0j}, b_{1j})^\top \sim N(0, D)$, inducing within-cluster dependence. The covariates are drawn from simple distributions and the fixed-effects component $f(\cdot)$ is modified across four scenarios to test model robustness under increasingly complex conditions.

The first scenario matches a correctly specified linear GLMM, providing a baseline. The second introduces a nonlinear effect through a logarithmic term; the third adds an interaction between predictors; the fourth generates strong multicollinearity via correlated covariates. These designs isolate challenges such as nonlinearity, interactions, and collinearity while preserving the same hierarchical structure.

All models are trained and evaluated using cluster-balanced 10-fold cross-validation, ensuring that folds preserve the grouping structure and preventing information leakage across clusters.

2.6 Evaluation metrics

Performance is evaluated using accuracy, F1 scores, and a simple prediction bias measure. Accuracy summarizes overall classification performance but can obscure differences in how models handle the event and non-event classes (Hastie et al., 2009). I therefore report two F1 scores—one for the majority class and one for the minority class—each defined as

$$F1 = \frac{2(\text{precision} \times \text{recall})}{\text{precision} + \text{recall}}$$

which provides a balanced measure of classification quality when precision–recall trade-offs differ across models.

To assess systematic over- or under-prediction, I compute a bias index

$$\text{Bias} = \frac{\sum \hat{y}}{\sum y} - 1.$$

Positive values indicate a tendency to over-predict the event, negative values the reverse.

3 Preliminary results

Table X reports mean F1 scores for each model across the four simulated scenarios. Several consistent patterns emerge.

In the **baseline** setting, where the fixed-effects structure is linear and correctly specified, the GLMM achieves the highest performance (0.840), as expected. GMERF closely follows (0.837), while GMERT also improves upon its single-level counterparts. Logistic regression, CART, and RF perform similarly. The small gap

between CART and RF reflects the low dimensionality of the predictor space: with only a few covariates, bagging and feature subsampling offer limited advantages, and random forests behave similarly to single trees.

In the **nonlinear** scenario, all models experience a performance drop. RF and CART show the largest decline, again exhibiting similar behaviour due to the limited number of predictors and shallow trees. In contrast, the multilevel machine-learning models retain much of their performance. GMERT remains close to the GLMM (0.814 vs. 0.822), indicating that explicitly modelling clustering substantially mitigates the loss induced by fixed-effect misspecification.

In the **interaction** scenario, multilevel corrections become particularly beneficial. Both GMERF (0.859) and GMERT (0.849) outperform their single-level tree and forest counterparts by a wide margin, approaching the GLMM benchmark. This highlights that much of the performance gain comes not from the tree structure alone, but from incorporating the hierarchical dependence through random effects.

Under **multicollinearity**, differences between single-level ML methods remain small, while multilevel models consistently dominate. GMERF and GLMM show the strongest robustness, whereas GMERT and single-level trees are more affected. Across all scenarios, introducing a multilevel correction systematically improves the performance of machine-learning models, often closing most of the gap with the parametric GLMM.

Taken together, the patterns observed across the simulated scenarios shed light on how machine-learning models behave in the presence of clustered data and on the role played by explicit multilevel corrections. Across all settings, single-level machine-learning methods tend to perform worse than their multilevel extensions, even when the predictor space is small and the data-generating process is relatively simple. Introducing random effects through GMERT and GMERF is associated with systematic improvements, suggesting that accounting for within-cluster dependence is a key driver of predictive performance. At the same time, differences between multilevel machine-learning models and the GLMM vary across scenarios, indicating that the benefits of flexible fixed-effect modelling depend on the structure of the underlying signal. These results motivate further investigation into the conditions under which multilevel machine-learning approaches provide substantive advantages, and into how they might be extended or refined in more complex or high-dimensional settings.

4 Next steps

4.1 Real data application

To complement the simulation study, the same modelling strategies will be applied to large-scale administrative data provided by Statistics Netherlands (CBS) on residential vacancies. This application serves to examine how the different approaches behave when applied to data that are not generated under controlled assumptions. The dataset includes a binary indicator of vacancy over a fixed period together with a large set of covariates describing dwelling characteristics, household composition, and socio-demographic attributes of residents, organised across multiple hierarchical levels. Classical mixed-effects models, single-level machine-learning methods, and their multilevel extensions will be compared within a common evaluation framework. The analysis will focus on how explicitly modelling clustering shapes predictive behaviour and model stability in a high-dimensional, real-world setting, and on how these patterns relate to those observed in the simulation scenarios.

4.2 More methods

The analysis can be extended by incorporating additional modelling approaches within a multilevel framework. Neural networks with hierarchical or random-effects components can be implemented to model complex nonlinear relationships while accounting for clustering (Zhang et al., 2021). Boosting-based mixed-effects models can be used to combine incremental function fitting with explicit random effects(Sigrist, 2024). Gaussian process models provide a nonparametric approach that naturally encodes dependence across levels(Sigrist, 2022). In high-dimensional settings, penalized mixed-effects models, such as LASSO extensions, will also be considered (Schelldorfer et al., 2014). While not a machine-learning method in a strict sense, they provide a natural alternative to standard regression when the number of predictors is large. ## More scenarios

A further extension concerns simulations in high-dimensional settings, which are particularly relevant for assessing the behaviour of multilevel methods when the number of predictors is large. These scenarios are not been implemented at this stage due to computational constraints. Future work may consider alternative simulation settings to explore how multilevel machine learning methods work under other conditions, like sparsity or different cluster dimensions.

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